



AMAN BHATT

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📄 Summary

Machine Learning Engineer with 4+ years of experience building production AI systems for power grids and renewable energy. Designed and deployed end-to-end demand forecasting platforms achieving <3% MAPE, production RAG applications using Llama 3.3 70B, and scalable ML infrastructure with FastAPI, MLflow, Docker, and AWS SageMaker.

👛 WORK EXPERIENCE

- Senior Machine Learning Engineer**, BSES Rajdhani Power Limited (BRPL) 📍 2025 Apr – Present
New Delhi
- Own end-to-end demand forecasting for Delhi's electricity grid — day-ahead, intraday, and week-ahead — at <3% MAPE, replacing a manual process that ran at 5–6%; segment-based ensembles (LightGBM, CatBoost, XGBoost) across weekday, weekend, and holiday regimes with real-time SCADA correction, Optuna tuning, and MLflow tracking.
 - Architected two production RAG systems on Llama 3.3 70B (ChromaDB, FastAPI, cross-encoder reranking): a forecasting assistant for grid operations, and a company-wide knowledge base over technical documentation and operational standards.
 - Built supporting data infrastructure — a FastAPI weather ingestion microservice across 14 geographic centroids feeding the forecast pipeline.
- Machine Learning Engineer**, Renew Power (Climate Connect Digital) 2022 Jun – 2025 Apr
Gurugram
- Built demand forecasting systems for three Delhi DISCOMs (BRPL, BYPL, TPDDL) from the ground up — segment-based ensembles (Neural Networks, LightGBM, CatBoost, XGBoost) across day-ahead and intraday horizons.
 - Designed solar and wind forecasting across 166 plants for REMC-Telangana; week-ahead predictions deployed on AWS SageMaker with automated CI/CD and model versioning.
 - Built electricity price forecasting models for IEX at 7.5% MAPE using LightGBM on market microstructure data.
- Engineer**, S & B Engineers & Constructors 2019 Aug – 2020 Sep
Noida
- Designed structural foundations for oil refinery equipment using AutoCAD, StaadPro, and RISA Base.

🧠 TECHNICAL SKILLS

Machine Learning & AI

Python | SQL | Statistics | Demand Forecasting |
Time-Series | Feature Engineering | LightGBM |
CatBoost | XGBoost | Scikit-learn | Optuna |
Ensemble Methods

Deep Learning

PyTorch | TensorFlow | Keras | ANN | CNN | RNN |
LSTM | GRU | Transformers | Autoencoders

Generative AI

RAG | LLM Integration | Llama 3.3 | ChromaDB |
LangChain | Vector Databases | Prompt
Engineering | NLP | LangGraph | Agentic AI

MLOps & Data Engineering

MLflow | AWS SageMaker | FastAPI | Docker |
Kubernetes | Airflow | CI/CD | Apache Spark |
Kafka | Pandas | NumPy | Linux | Git

🎓 EDUCATION

B.Tech, Civil Engineering, Aligarh Muslim University 📍 2015 – 2019
Grade: 9 CGPA